INCH-POUND
MIL-DTL-16377/26B(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/26A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 110 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 112, 113, 113.1, 112SSL, 113SSL, AND 113.1SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

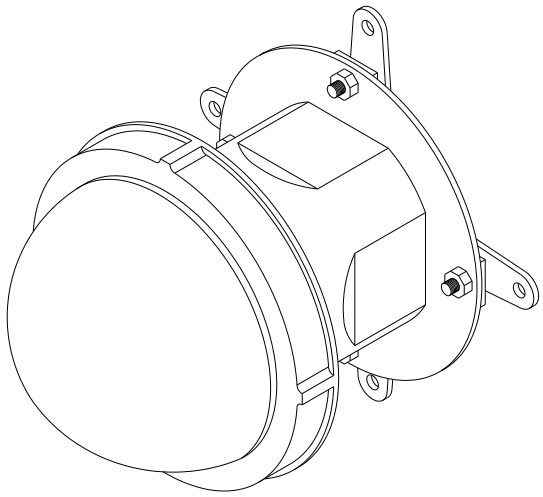


FIGURE 1. Fixtures, incandescent, symbols 112 and 113.

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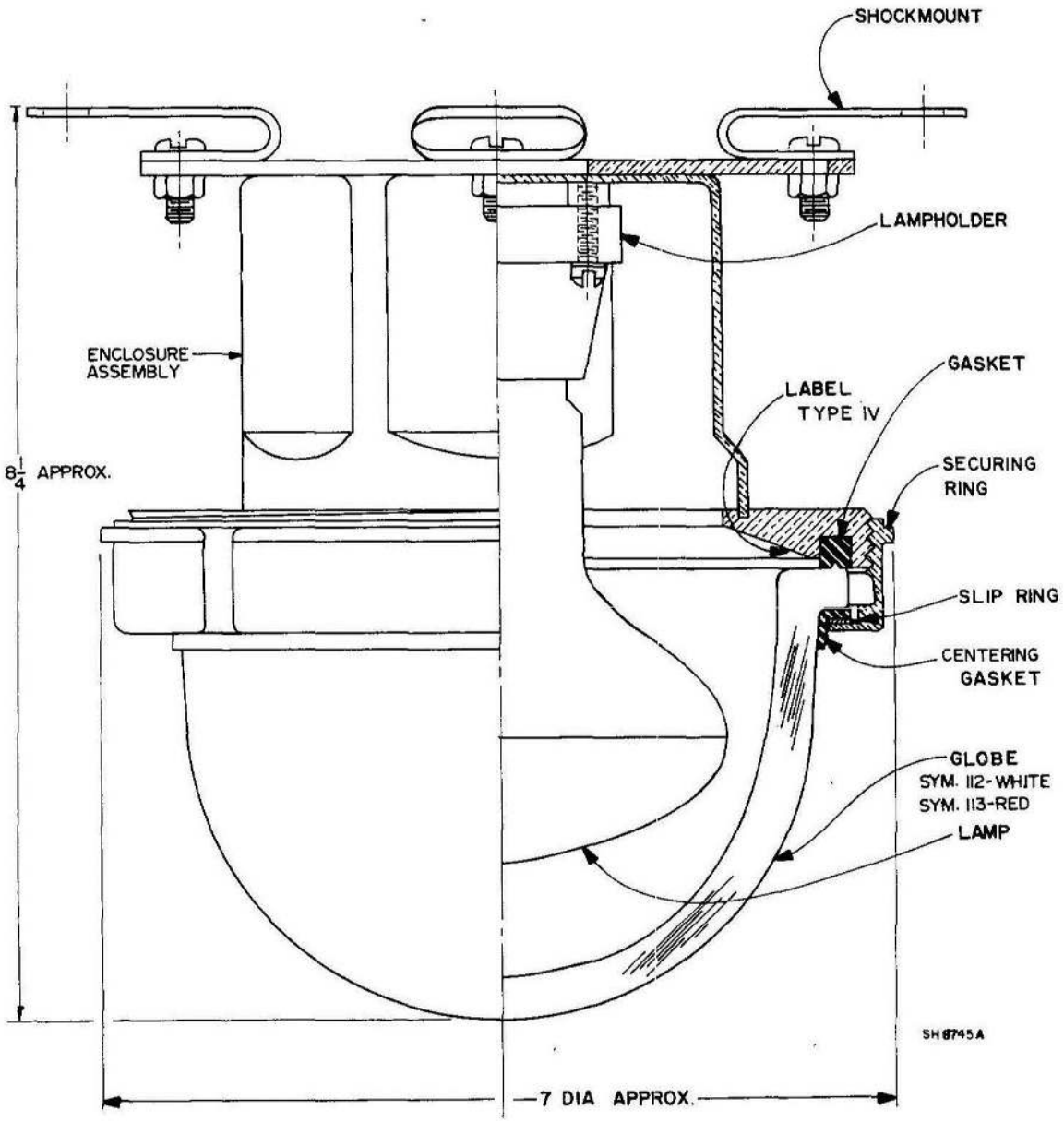


FIGURE 2. Fixture assembly.

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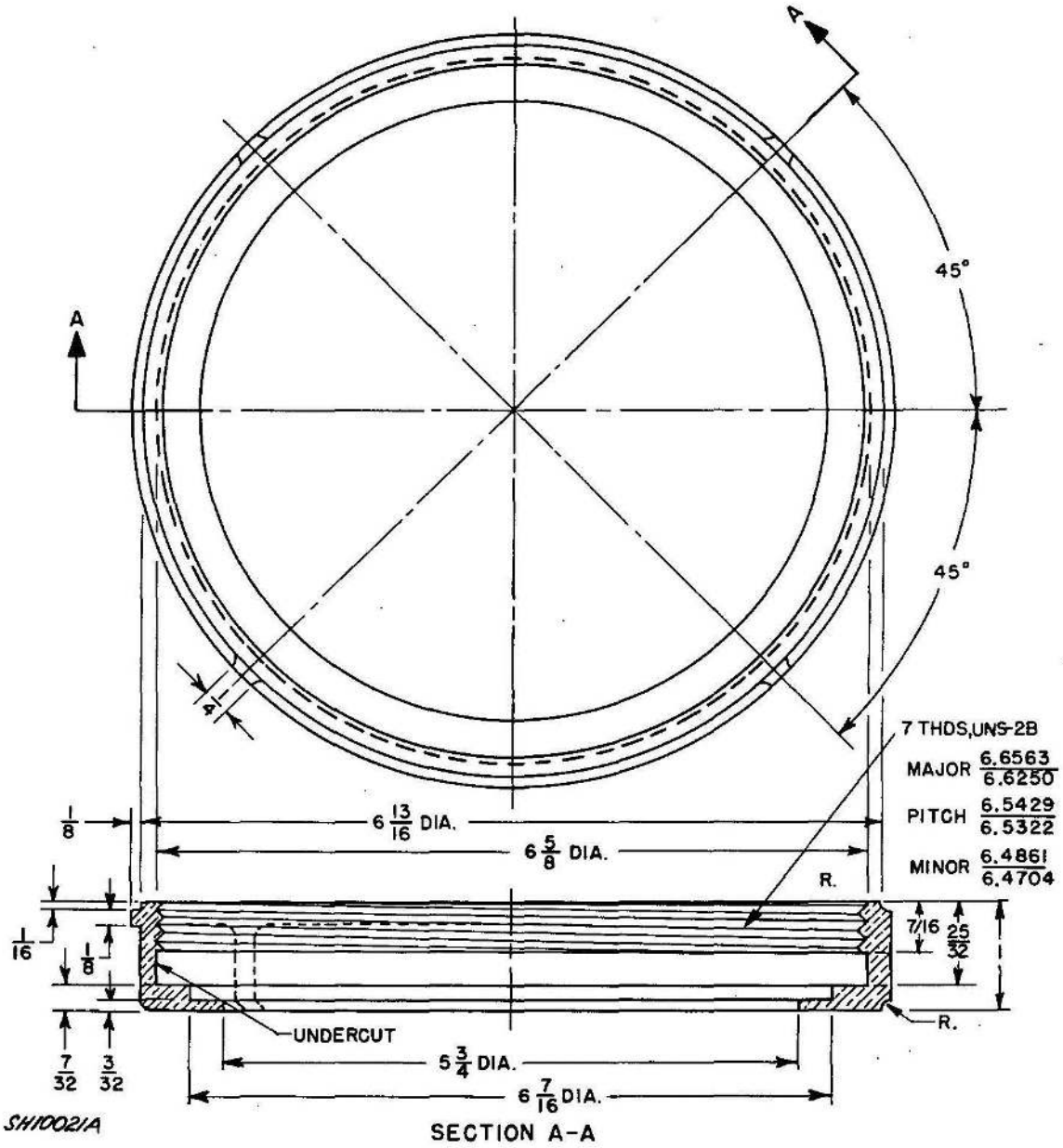


FIGURE 3. Securing ring.

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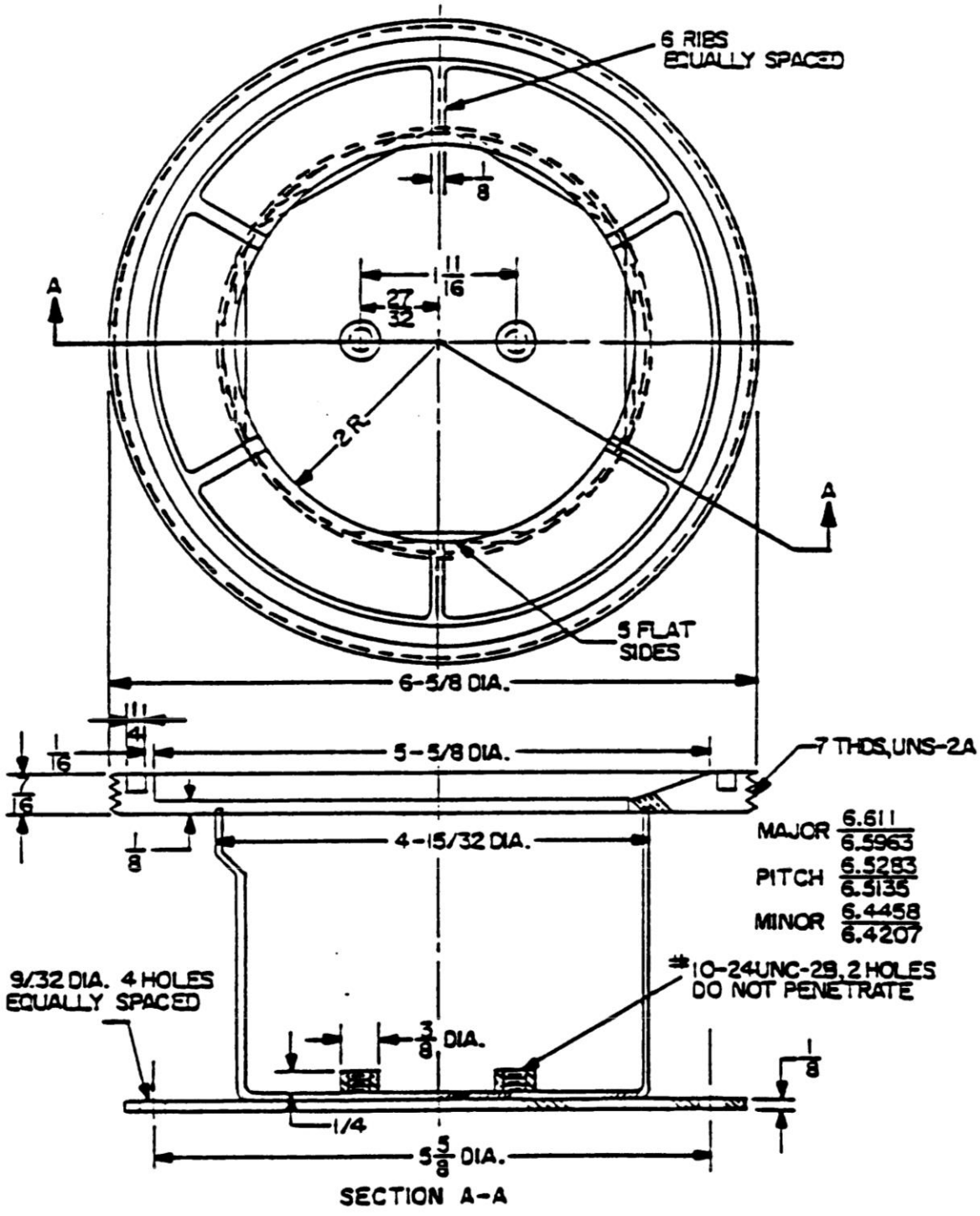


FIGURE 4. Enclosure assembly.

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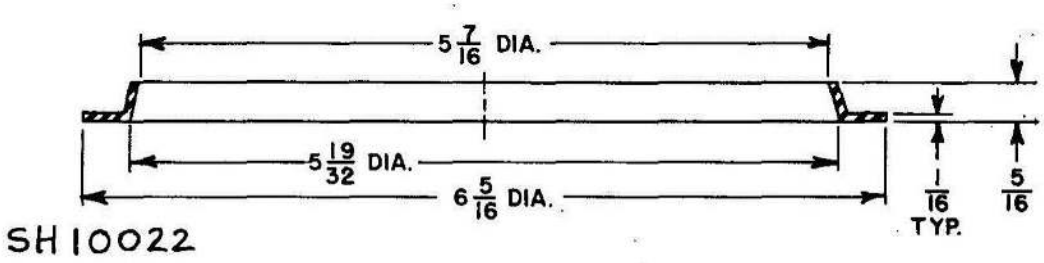


FIGURE 5. Globe centering gasket.

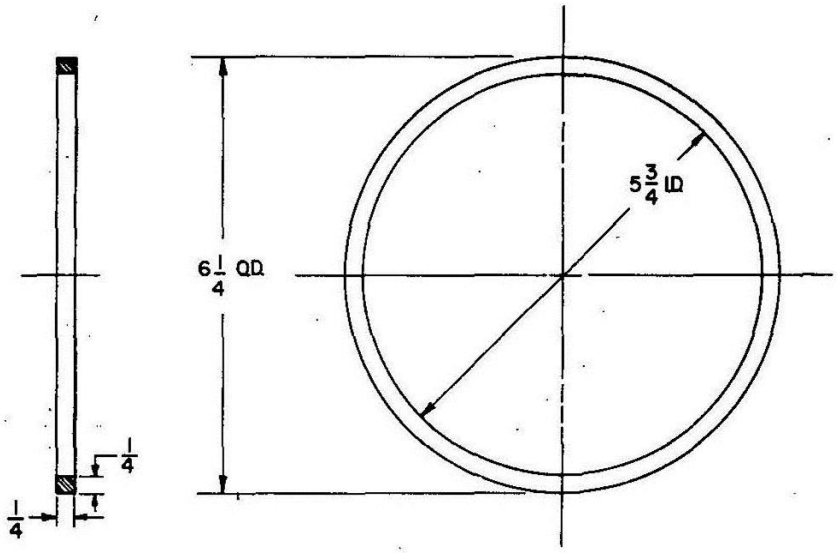


FIGURE 6. Enclosure gasket.

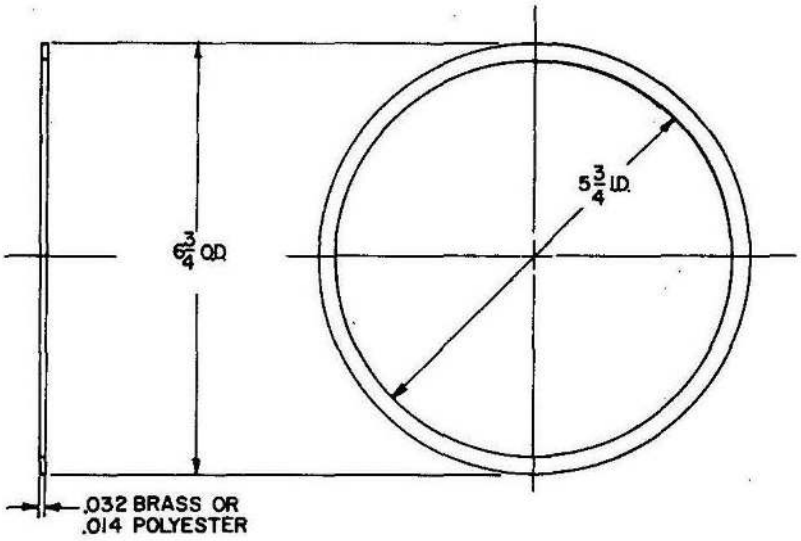
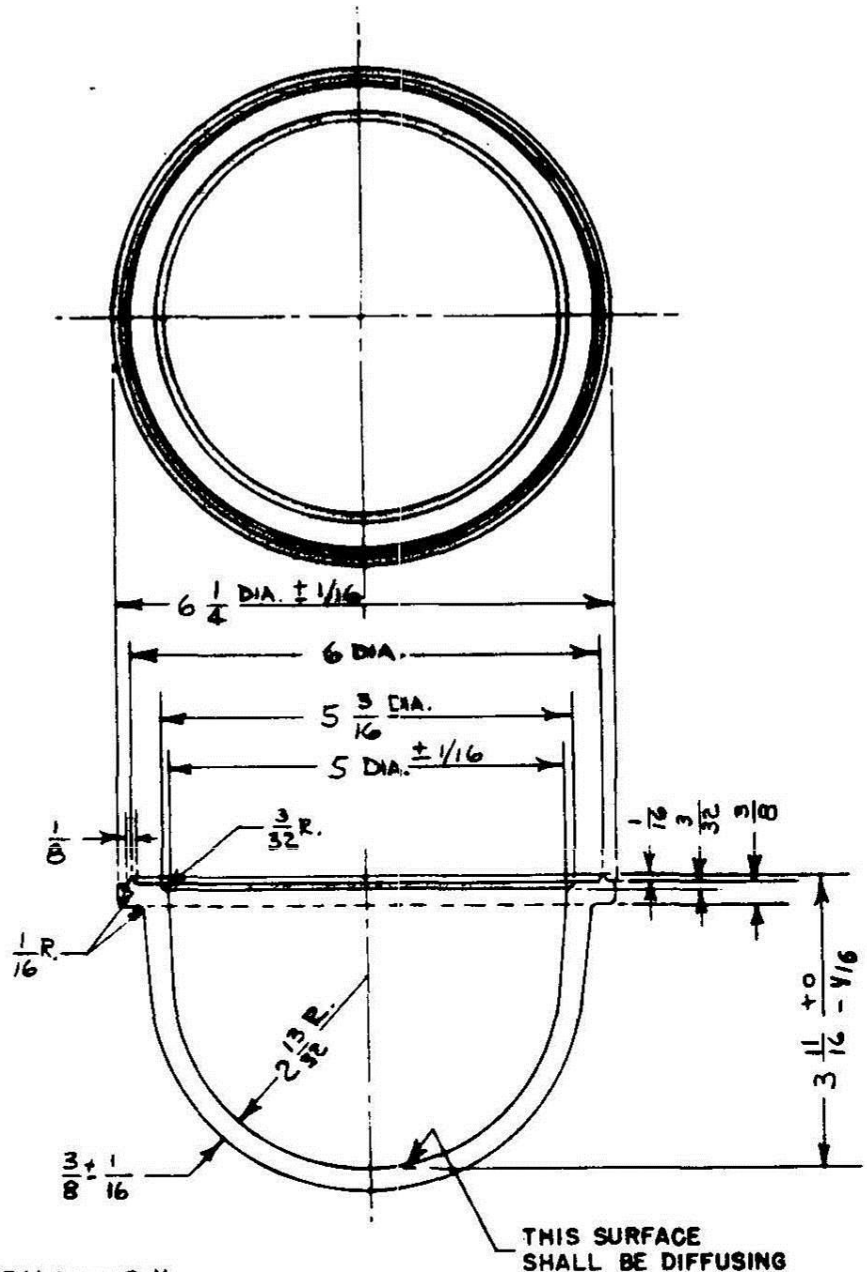


FIGURE 7. Slip ring.

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Material: High impact, heat-resistant glass.

Color:

Symbol 112: White (clear) transparent/diffusing.

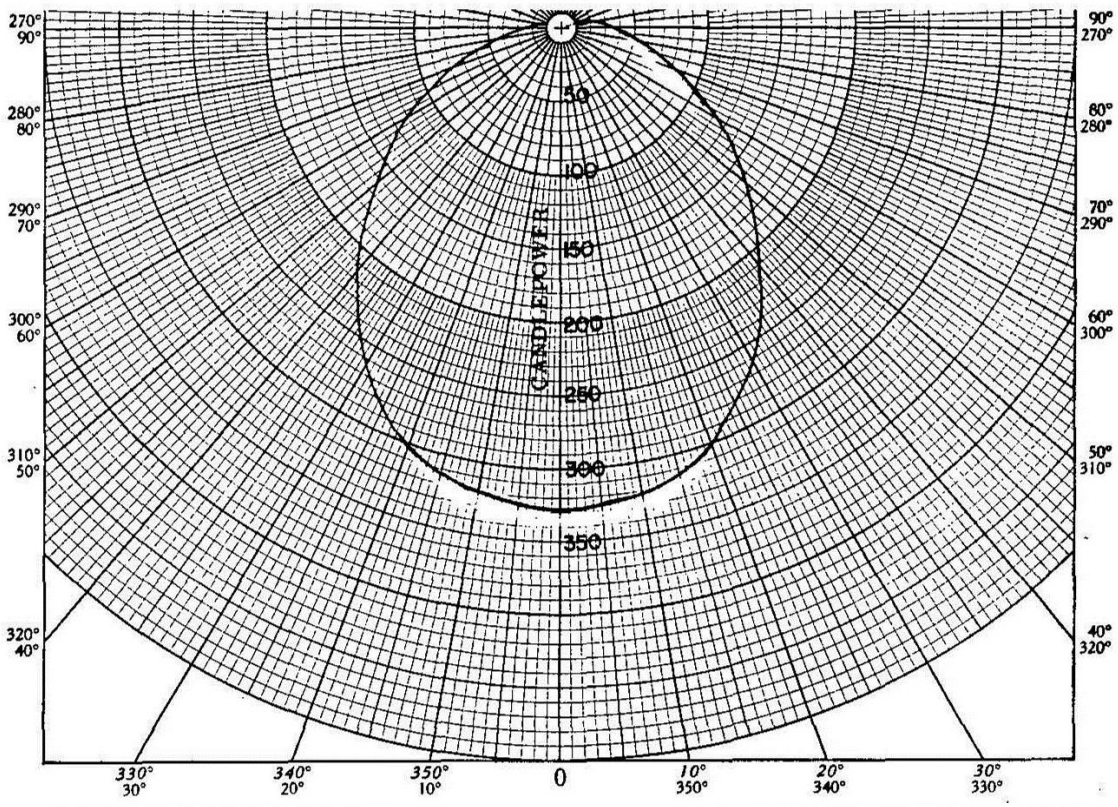
Symbol 113: Red transparent/diffusing.

Symbol 113.1: Yellow transparent/diffusing.

Transmittance (Symbol 112): Clear transparent/diffusing – 80 percent minimum.

FIGURE 8. Globe.

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ANGLE	CANDLEPOWER SYMBOL 112
0	330
5	320
10	318
15	315
20	300
25	280
30	260
35	240
40	214
45	185
50	163
55	145
60	125

ANGLE	CANDLEPOWER SYMBOL 112
65	105
70	85
75	55
80	50
85	40
90	35
95	25
100	20
105	15
110	11
115	9
120	7
125	6

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Voltage:	120 volts.
Ambient temperature:	76 °F (24.4 °C).
Lamps:	110 watts, rough service conforming to MS16563-1, furnished by laboratory for test, shall be used.

FIGURE 9. Candlepower distribution curves.

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INCANDESCENT REQUIREMENTS:

CLASSIFICATION:

Type: II
Class: 2

SYMBOL NUMBERS:

112: Complete lighting fixture assembly, general lightning, with one 110-watt white lamp and white (clear transparent) diffusing globe.

113: Complete lighting fixture assembly, general lighting, with one 110-watt white lamp and red transparent diffusing globe.

113.1: Complete lighting fixture assembly, general lighting, with one 110-watt white lamp and yellow transparent diffusing globe.

DESIGN:

Electrical rating:	110 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 through 9.
Enclosure assembly:	Brass, see figure 4.
Gaskets:	Rubber, ¼ inch square cross section by 5¾ inside diameter (id) by 6¼ outside diameter (od).
Lampholder:	UL 496, SUPP. SA, FIG. SA3.6, two-contact.
Shock mounts:	MIL-F-16377/41-007.
Mounting dimensions:	See figure 28 of MIL-DTL-16377.
Globes:	
Symbol 112:	White (clear transparent), see figure 8.
Symbol 113:	Red, see figure 8.
Symbol 113.1:	Yellow, see figure 8.
Centering gasket:	Molded rubber, see figure 5.
Slip ring:	Brass, 6¼ od by 5¾ id, by 1/32 inch thick.
Securing ring:	Cast brass or bronze, see figure 3.
Lamp:	Legacy incandescent fixtures shall be designed to accommodate a 110-watt, white lamp in accordance with MS16563-1. (Lamp shall not be furnished.)
Weight (Symbols 112 and 113):	8 pounds 9 ounces maximum.
Distribution curves (Symbol 112):	See figure 9.
Designation and marking:	Labels shall be furnished in accordance with MIL-DTL-16377. One Type IV label shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be "USE 110-WATT LAMP ONLY LAMP: NSN 6240-00-578-6820 WHITE GLOBE: NSN 6210-00-635-8819 RED GLOBE: NSN 6 21 0-00-635-8820"

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

112SSL: Complete lighting fixture assembly, general lightning, with LED light source and white (clear transparent) diffusing globe.
113SSL: Complete lighting fixture assembly, general lighting, with LED light source and red transparent diffusing globe.
113.1SSL: Complete lighting fixture assembly, general lighting, with LED light source and yellow transparent diffusing globe.

DESIGN:

Electrical rating: 110 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration: See figures 1 through 9.
Photometric data: See figure 9 and 3.7.1 of MIL-DTL-16377.
Chromaticity: See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit: See 3.17 of MIL-DTL-16377.
Mounting dimensions: See figure 28 of MIL-DTL-16377.
Enclosure: Watertight. See 3.5.11 of MIL-DTL-16377.
Globes:
Symbol 112SSL: White (clear transparent), see figure 8.
Symbol 113SSL: Red, see figure 8.
Symbol 113.1SSL: Yellow, see figure 8.
Weight (Symbols 112 and 113): 8 pounds 9 ounces maximum.
Designation and marking: Labels shall be furnished in accordance with MIL-DTL-16377.

a. One Type IV label shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by:
(1) A list of all standard parts shown in table IV and globes of table IV by name and stock number only, if appropriate, or
(2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table I for Type I and table II for Type III.
Globes: See table III.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table IV.

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TABLE I. First article and comparison inspection for type I fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.7.1	4.8.4	
Mechanical shock	3.7.7.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.7.7	4.8.17.4	
Breakage pattern	3.7.7.9	4.8.17.6	
Color and luminous transmittance	3.7.3	4.8.6.3	Symbol 113 and 113.1 only
Light output	3.7.1	4.8.6.1	Symbol 112 only Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE II. First article and comparison inspection for type III luminaires. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.14	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Mechanical shock	3.7.7.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.7.7	4.8.17.4	
Breakage pattern	3.7.7.9	4.8.17.6	
Color and luminous transmittance	3.7.3	4.8.6.3	Symbol 113SSL and 113.1SSL only
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Salt spray	3.5.15	4.8.10	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE III. First article and comparison inspection for globes.

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.7.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 112 only Candlepower
Mechanical shock	3.7.7.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.7.7	4.8.17.4	
Breakage pattern	3.7.7.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 113 only

NOTE:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

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TABLE IV. Part number and general information.

Military Part Number M16377/26-	Item Name	National Stock Number (NSN)	Remarks
112	Complete fixture, Symbol 112 consisting of: Body Clear transparent diffusing globe	 6210-00-635-8817 6210-00-635-8819	
113	Complete fixture, Symbol 113 consisting of: Body Red transparent diffusing globe	 6210-00-635-8817 6210-00-635-8820	
113.1	Complete fixture, Symbol 113.1 consisting of: Body Yellow transparent diffusing globe	 6210-00-635-8817 6210-01-166-7185	
112SSL	Complete fixture, solid state, Symbol 112SSL		
113SSL	Complete fixture, solid state, Symbol 113SSL		
113.1SSL	Complete fixture, solid state, Symbol 113.1SSL		
001	Legacy fixture body (minus globe)	6210-00-635-8817	
002	White (clear transparent diffusing) globe	6210-00-635-8819	1/
003	Red (transparent diffusing) globe	6210-00-635-8820	2/
004	Securing ring (legacy incandescent only)	6210-00-399-6208	3/
005	Yellow (transparent diffusing) globe	6210-01-166-7185	4/

NOTES:
1/ Formerly covered on MS17802-10.
2/ Formerly covered on MS17801-11.
3/ Supersedes part number 9000-S6401-74219-1.
4/ Formerly referred to as M16377/26-003 MODIFIED.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-029)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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